
Governance Interlock Protocol (G.I.P.): Definition and Implementation of a Dynamic Pre-Generation Condition Confirmation Model for Probabilistic Constraint Compliance in Large Language Models

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Abstract

This paper proposes and formalizes a novel prompt engineering methodology, the **Governance Interlock Protocol (G.I.P.)**, which probabilistically improves the rate of constraint compliance in large language model (LLM) output generation through a *pre-generative* and *intrinsic* approach: requiring the model to perform a linguistic self-attestation of the integrity of its execution environment immediately prior to response generation. LLMs are probabilistic token generation systems; even after alignment training, references to constraints placed early in the prompt weaken as session length grows due to attention decay, causing settings such as persona, bias control, and hallucination suppression to drift (Benderahmane et al., 2024). Whereas existing countermeasures address this problem from the direction of post-hoc correction, censorship, or re-injection, G.I.P. operates on a different design philosophy: it requires the model to linguistically self-verify the functional state of pre-defined preconditions immediately before response generation, then hands off the resulting output to a subordinate audit protocol, thereby raising the probability of constraint compliance from the point of generation itself.

The originality of G.I.P. lies in a paradigm shift toward a "pre-generative, intrinsic approach," fundamentally distinct from: (i) the limitations of self-correction without external feedback identified by Huang et al. (2024); (ii) the training-time intervention adopted by the instruction hierarchy research of Wallace et al. (2024); and (iii) the post-hoc critique-and-revision cycle proposed by Constitutional AI (Bai et al., 2022).

The core finding this paper most emphasizes is the **extensibility** of G.I.P. Beyond the standard three conditions of *assumed user profile*, *persona*, and *bias configuration*, incorporating any condition items chosen at the prompt designer's discretion into the pre-confirmation sequence has been observed to improve compliance with those conditions and to suppress drift. Furthermore, this extensibility manifests even when the post-hoc audit protocol has been removed. In other words, G.I.P. functions as an independent control paradigm—inherent to the pre-confirmation mechanism itself—capable of arbitrary

extension regardless of the content or type of conditions being confirmed, and without dependence on any audit function.

This paper systematically formalizes G.I.P.'s theoretical framework, its differentiation from existing methods, its implementation as a multi-layered defense architecture, and its condition confirmation sequence and flow. It reports the protocol's effectiveness based on empirical observations from its implementation in a practical investment analysis prompt requiring at least eight consecutive turns of output.

Keywords: Governance Interlock Protocol; pre-generation condition confirmation; point-of-generation control; persona drift prevention; self-attestation mechanism; multi-layered defense architecture; re-prompting; instruction hierarchy; constraint compliance probability; prompt engineering

1. Introduction

The core challenge in operating large language models (LLMs) is not the question of judging the quality of outputs, but rather the problem that the model itself forgets or under-weights whether the specified constraint conditions are *currently in effect*. LLMs are essentially probabilistic token generation systems and cannot be completely controlled by prompts; however, statistically improving the probability of constraint compliance is achievable as a matter of design theory.

The phenomenon whereby an alignment-trained LLM exhibits weakening references to early prompt instructions as the session grows longer has been quantitatively measured by Benderahmane et al. (2024) and formalized as **attention decay**, identified as the principal cause of drift in persona and rule settings. Existing research has primarily addressed this problem from three directions.

The first is the direction of post-hoc **self-correction**: a symptomatic approach that searches for and corrects errors after generation. As Kamoi et al.'s (2024) critical survey demonstrates, the improvements claimed by prior work on intrinsic self-correction—self-correction without external feedback—do not reproduce, and Huang et al. (2024) empirically demonstrated the difficulty of LLMs self-correcting their reasoning without external feedback. There is a fundamental problem of unavoidable anchoring effects toward the already-generated erroneous answer.

The second is the direction of **instruction hierarchy** design. Wallace et al. (2024) proposed a method of training LLMs to prioritize higher-priority instructions. However, this approach requires training-time intervention via fine-tuning and does not envisage implementation as a design theory operating through prompt text alone.

The third is the direction of control via natural-language principle sets, exemplified by **Constitutional AI** (Bai et al., 2022), which operates on the design philosophy of having the model critique and revise its own outputs against a set of principles. This too, however, is designed as a post-hoc revision cycle and lacks the perspective of intervention at the point of generation.

This paper proposes a fourth direction, distinct from these existing methods: the design philosophy of inserting a *pre-generative, intrinsic* condition-confirmation mechanism at the point of generation. We name this mechanism the **Governance Interlock Protocol**

(G.I.P.) and provide its theoretical formalization together with a practical implementation report.

The core conception of G.I.P. is to direct the model's attention preferentially to the relevant conditions by having it linguistically articulate, immediately prior to response generation, the functional state of the pre-defined preconditions. This is not a process of "seeking the correct answer," but rather an operation of confirming in advance the integrity of the execution environment—whether the model is in a *state in which a correct answer can be produced*. This can be understood as a design intervention that conforms to the operating principles of the attention mechanism presented by Vaswani et al. (2017).

2. Background and Related Work

2.1 Attention Decay and the Drift Problem in LLMs

In the Transformer architecture introduced by Vaswani et al. (2017), the attention mechanism assigns weights to each token in the context and influences the generation of subsequent tokens based on those weights. As a session lengthens, the number of tokens in the context window increases, and the relative attention weight directed at the system-setting tokens placed early in the prompt declines—a phenomenon termed *attention decay*.

Benderahmane et al. (2024, arXiv:2402.10962, COLM 2024) quantitatively measured this phenomenon as **persona drift**, showing that personas degrade markedly within 8 to 12 turns in long-form dialogue. While that work proposes *split-softmax* as a post-hoc suppression technique, it does not address intervention at the point of generation through prompt text alone.

Choi et al. (2025, arXiv:2412.00804) examined identity drift in nine LLMs, showing that drift is greater for larger models and that persona assignment alone is insufficient for drift prevention. This suggests that, in addition to static configuration descriptions, a dynamic reference mechanism is required.

Li et al. (2025, arXiv:2510.07777) presented the observation that drift does not accumulate without bound, suggesting that lightweight interventions can bring drift toward an equilibrium state. G.I.P. can be positioned as one form of such "lightweight intervention," but differs fundamentally in that the intervention is generated dynamically at the point of generation in each turn.

2.2 Differentiation from LLM Self-Correction Research

The critical survey by Kamoi et al. (TACL 2024, arXiv:2406.01297) showed that no prior work demonstrates successful self-correction by prompted LLMs using their own feedback (with the exception of a small number of tasks particularly suited to self-correction); that self-correction is effective when reliable external feedback is available; and that large-scale fine-tuning enables self-correction.

Huang et al. (ICLR 2024) empirically demonstrated the difficulty of LLMs self-correcting reasoning without external feedback, reporting numerous cases in which performance degraded after self-correction. This finding is positioned as an important antecedent that

clarifies G.I.P. is *not* designed as self-correction. G.I.P. does not revise generated outputs; it functions as preprocessing that confirms the functional state of conditions before generation.

2.3 Differentiation from Constitutional AI

Constitutional AI (CAI) by Bai et al. (2022) is a method that designs a cycle in which the model critiques and revises its own outputs against a set of natural-language principles, functioning primarily as alignment control during training. Huang et al. (CCAI, FAccT 2024) extended this framework to a public participation process.

G.I.P. and CAI share the design direction of "explicit principles in natural language," but G.I.P. is positioned as an independent design theory in two respects: (i) it functions through prompt text alone, requiring no training-time intervention; and (ii) it operates as pre-generation confirmation at the point of generation, rather than as a post-hoc critique-and-revision cycle.

2.4 Differentiation from Instruction Hierarchy Research

The instruction hierarchy research of Wallace et al. (2024, arXiv:2404.13208) proposed a method of training LLMs to prioritize system prompts, substantially improving robustness against prompt injection and similar attacks. "Reasoning Up the Instruction Ladder" (2025, arXiv:2511.04694) proposed an approach that reformulates compliance with the instruction hierarchy as a meta-reasoning task explicitly performed before executing the user request.

Both of these research directions require training-time intervention or fine-tuning. G.I.P. requires neither architectural modification nor fine-tuning; it operates on a different design dimension by causing condition confirmation to occur at the point of generation in each turn through natural-language text within the prompt alone.

2.5 Summary of Differentiation from Prior Work

Synthesizing the differentiation from the above body of prior work: a design theory that, immediately prior to generation, uses prompt text alone to have the model linguistically execute a self-attestation of preconditions (assumed user profile, persona, bias configuration, and so forth)—and thereby probabilistically improves constraint compliance—has not been found in the prior literature. Furthermore, the empirical finding that this can be achieved with arbitrary extension—incorporating any conditions, regardless of content or type, into the pre-confirmation sequence and observing improved compliance with each, even in the absence of an audit protocol—has likewise not been found in the prior literature. G.I.P. is positioned as a design that fills this gap.

3. Theoretical Framework of G.I.P.

3.1 Basic Definition

G.I.P. is a dynamic condition-confirmation mechanism in which, one step before generating a response, an LLM (i) self-attests that pre-defined preconditions have been correctly loaded and are functioning, (ii) executes generation in that state, and (iii) hands off the generated response to a subordinate audit protocol.

This attestation differs fundamentally from a process of "seeking the correct answer." It is a technique for directing attention preferentially by having the model linguistically articulate—through self-description—the integrity of the execution environment, that is, *whether the model is in a state in which a correct answer can be produced*. It functions as explicit re-activation of conditions and is positioned as a preventive intervention at the point of generation against the problem of attention decay formalized by Benderahmane et al. (2024).

In its standard implementation, G.I.P. takes three conditions as the targets of confirmation—**assumed user profile**, **persona**, and **bias configuration**—but this does not mean that G.I.P.'s confirmation targets are fixed at three. The basis for selecting these three as the standard is that they function as the foundational generative conditions that fundamentally regulate the nature of responses. The *assumed user profile* defines the prerequisites for the response's evaluative axes, weightings, and scenario evaluation; the *persona* regulates the level of specialized knowledge and viewpoint; and the *bias configuration* secures the objectivity of evaluation by prohibiting unjustified accommodation, affirmation, and neutrality. In investment analysis, for example, setting the time horizon based on the user's assumed investment period, securing the level of description through a financial-analytic specialized viewpoint, and maintaining objectivity unwarping by bias are indispensable preconditions for the description of every analytic item; generation in a state where these are absent damages the validity of the entire analysis. In this sense, these three conditions are standardized as foundational conditions that must be secured at the point of generation, prior to subsequent detailed scrutiny.

On the other hand, the fundamental property of G.I.P. is its **extensibility**: regardless of the content or type of conditions incorporated into the pre-confirmation sequence, compliance with those conditions improves probabilistically. Empirical observation has confirmed that any condition the prompt designer wishes to maintain can be added to the pre-confirmation sequence; attention to that condition is dynamically regenerated at the point of generation in each turn, functioning as drift suppression.

It should be noted that G.I.P.'s self-attestation acts upon the model's internal probabilistic processing and does not guarantee deterministic control at the hardware or software level. G.I.P. functions as a shift in the probability distribution and aims at probabilistic improvement of constraint compliance.

3.2 The Decisive Difference from Existing Verification and Audit Prompts

G.I.P.'s originality lies in a paradigm shift from the conventional "post-hoc, external" approach to a "pre-generative, intrinsic" approach.

Conventional verification and audit (post-audit/censorship) is symptomatic work that searches for errors after generation. As Kamoi et al. (2024) point out, it has the fundamental problem of being unable to avoid anchoring effects toward the already-generated erroneous answer.

In contrast, G.I.P. is preemptive control that prompts linguistic re-confirmation of settings before response generation. By having the operator perform a "point-and-call" check on the production line's state immediately before flipping the switch, it raises the probability of constraint compliance from the point of generation itself. This is positioned as a third design theory, distinct from both the "post-hoc self-correction" critically examined by Huang et al.

(2024) and the "training-time instruction-hierarchy assignment" proposed by Wallace et al. (2024).

4. Multi-Layered Defense Architecture: G.I.P. + Audit Protocol

4.1 The Principle of Robustness Reinforcement

The true value of G.I.P. is realized when an existing self-audit prompt is incorporated into its lower layer. Mere self-audit functions poorly when the model has forgotten its constraints. However, by having G.I.P. linguistically re-confirm that "the three foundational conditions of assumed user profile, persona, and bias configuration are active," and then handing off the resulting output to the audit protocol, the robustness of governance improves statistically.

In this multi-layered defense model:

- **Re-prompting (Layer 1)** functions as the foundation that re-activates referential attention to system settings before G.I.P. is executed.
- **G.I.P. (Layer 2)** functions as upper-tier control that preferentially directs attention to the state in which "the foundational conditions of generation are being applied," and then executes response generation.
- **Audit protocol (Layer 3)**, receiving that output, functions as the practical audit that "scrutinizes the details of the generated content."

Structurally, the division of roles can be formalized as follows:

- **Re-prompting (Layer 1):** At the head of each turn, re-declare the principal control rules and re-activate attention.
- **G.I.P. (Layer 2):** Point-of-generation confirmation of the three conditions (assumed user profile, persona, bias configuration) → response generation → handoff to the audit protocol.
- **Audit protocol (Layer 3):** Scrutinize and revise the response generated under the conditions secured by G.I.P., for compliance with detailed conditions such as hallucination suppression, computational execution principles, and information-use stipulations.

It should be noted that all three layers are subject to the influence of attention decay associated with context length, as described by Benderahmane et al. (2024); in extremely long sessions, the effectiveness of each layer may decline.

4.2 The Effectiveness of G.I.P. as a Standalone Operation

In the implementation observations on the investment analysis prompt described later, it was confirmed that G.I.P. alone—even with the audit protocol removed—was executed across all turns without drift in continuous analyses of at least eight turns holding multiple constraint conditions. This is positioned as an empirical record showing that G.I.P. is effective not only in the context of multi-layered defense, but also as a standalone probabilistic improvement of constraint compliance.

5. Implementation: The Condition Confirmation Sequence and Flow

5.1 Logical Definition by Pseudocode

The logical structure of G.I.P. is defined by the following pseudocode.

Governance Interlock Protocol:
Begin pre-generation process

Step 1: Assumed User Profile Condition Confirmation
(Confirm whether the assumed user profile is functioning as a generative condition for the response's evaluative axes and weightings.
If not applied, re-apply and proceed to Step 2.)

Step 2: Persona Condition Confirmation
(Confirm whether the persona configuration is functioning as a generative condition. If not applied, re-apply and proceed to Step 3.)

Step 3: Bias Configuration Condition Confirmation
(Confirm whether all conditions of the bias-suppression instructions are functioning as generative conditions. If not applied, re-apply and proceed to the response generation phase.)

↓

Response generation phase:
After confirming satisfaction of all conditions in Steps 1–3, generate and output the final response.
Display "Response Generation Condition Confirmation Flow Applied .

↓

Handoff to audit protocol:
Hand off the generated response to the independent subordinate audit protocol. (Detailed scrutiny of hallucination suppression, computational execution principles, information-use stipulations, etc., is performed at this layer.)

G.I.P. has the structure of executing response generation only after completing the three-stage condition confirmation in Steps 1–3, then handing off the output to the subordinate audit protocol. While G.I.P. itself takes as standard the confirmation of three foundational generative conditions—assumed user profile, persona, and bias configuration—its design is extensible: arbitrary condition items may be added to the Step group.

5.2 Linguistic Application Example (Condition Confirmation Sequence)

A standard linguistic implementation example of the condition confirmation sequence in G.I.P. is shown below.

"Immediately before output generation, strictly execute the following thought-condition confirmation flow. Omission of any step is forbidden.

>

Step 1: Assumed User Profile Condition Confirmation. If applied, proceed to Step 2. If not applied, re-apply the condition and proceed to Step 2.

>

Step 2: Persona Condition Confirmation. If applied, proceed to Step 3. If not applied, re-apply the condition and proceed to Step 3.

>

Step 3: Bias Configuration Condition Confirmation. If all conditions are applied, proceed to response generation. If any conditions are not applied, re-apply them and then proceed to response generation.

>

Response generation phase: Display 'Response Generation Condition Confirmation Flow Applied ' and output the final response. Then hand off to the audit protocol."

By means of this construction, the functional state of the three foundational generative conditions—assumed user profile, persona, and bias configuration—is linguistically confirmed before generation in each turn. Conditions that are not functioning are re-activated at the moment of articulation, producing recovery from the forgetting state caused by attention decay. After generation, the output is handed off to the subordinate audit protocol, where detailed scrutiny of content is performed under the conditions G.I.P. has secured.

5.3 Design Principles for Attestation Steps

In designing the condition confirmation sequence, the attestation steps follow the principles below.

Arbitrary extensibility of conditions: The standard implementation of G.I.P. takes the three conditions of assumed user profile, persona, and bias configuration as confirmation targets, but these three are not exhaustive of the conditions under which G.I.P. functions effectively. A fundamental property of G.I.P. is the extensibility whereby compliance probability improves regardless of the content or type of conditions incorporated into the pre-confirmation sequence. By adding to the sequence any condition the prompt designer wishes to maintain—hallucination suppression, computational execution principles, output formatting, information-use stipulations, and so forth—drift-suppression effects on those conditions emerge.

Reloading of the environment: Rather than asking *whether the model knows the instructions*, have it linguistically articulate, at each step, whether *that mode is currently ON*, thereby preferentially directing the model's attention to the relevant condition.

Stepwise confirmation: By having the confirmation items of each step be explicitly articulated in language, achieve probabilistic improvement of condition referencing in the subsequent generation phase. This acts not as a true software-level conditional branch or interrupt, but as attention guidance through the model's self-description.

Explicit handoff to the audit protocol: Once response generation under the conditions secured by G.I.P. is complete, a structure of explicit handoff of the generated artifact to the subordinate audit protocol clarifies the division of roles between G.I.P. (pre-confirmation)

and the audit protocol (post-hoc scrutiny). However, the effectiveness of G.I.P. does not presuppose the existence of the audit protocol. G.I.P. has independent effectiveness on its own: compliance with conditions included in the pre-confirmation sequence improves.

6. Implementation in an Investment Analysis Prompt and Observations

6.1 Overview of the Implementation Target

To verify the effectiveness of G.I.P., the implementation target chosen was an investment analysis prompt (Ver3) that, given only three input items—company name, ticker, and listing market—autonomously executes a four-chapter structure consisting of financial analysis and valuation analysis (Chapter 1), business and competitive analysis (Chapter 2), technical analysis (Chapter 3), and supplementary cautionary items (Chapter 4) over at least eight consecutive turns.

This prompt has the following composite constraint conditions: persona configuration (senior investment analyst); multi-layer bias control (prohibition of unjustified accommodation, affirmation, neutrality, and criticism); hallucination suppression (mandatory ⚠️ display when data is insufficient, mandatory explicit marking of inferences); computational execution principles (Python required); prohibition of the use of social media and bulletin-board information; processing flow for non-public information; strict observance of source-display stipulations; among many other independent constraint conditions whose simultaneous functioning is required by the design.

A multi-constraint, long-session prompt of this kind constitutes the condition under which the drift caused by attention decay—as shown by Benderahmane et al. (2024)—appears most prominently, and is therefore an appropriate environment for empirically demonstrating the effectiveness of G.I.P.

6.2 Implementation Structure of G.I.P.

The implementation of G.I.P. in the investment analysis prompt is designed as Section H ["Thought Verification Section"], explicitly labeled "Governance Interlock Protocol." The Step structure of the condition confirmation that G.I.P. handles is shown below.

- **Step 1:** Assumed User Profile Condition Confirmation (whether the assumed user profile is functioning as a generative condition for the response's evaluative axes, weightings, and scenario evaluation).
- **Step 2:** Persona Condition Confirmation (whether the persona configuration is functioning as a generative condition).
- **Step 3:** Bias Configuration Condition Confirmation (whether all conditions of the bias-suppression instructions are functioning as generative conditions—prohibition of unjustified accommodation, affirmation, neutrality, criticism, and so on).
- **Response generation phase:** Confirm satisfaction of Steps 1–3, display "Response Generation Condition Confirmation Flow Applied ✅," and output the final response.
- **Handoff to audit protocol:** Hand off the generated response to the self-audit phase of Section J ["End-of-Turn Execution Block"].

Steps 1–3 are executed before generation in each turn; if any condition is not functioning, it is re-applied before proceeding to the next step. Confirmation of detailed conditions—hallucination suppression, computational execution principles, stipulations on the use of social media and bulletin-board information, and so forth—is performed at the audit protocol layer (described later as Layer 3) after handoff from G.I.P. A monitoring mechanism (Section C ["Session Resumption Detection"]) is also implemented: if the Step-completion marker is absent from the output of the immediately preceding turn, session resumption is detected and re-execution is triggered.

6.3 Overall Design as a Multi-Layered Defense Architecture

The governance design in the investment analysis prompt is implemented as the multi-layered defense architecture proposed in this paper. Three independent mechanisms—attention re-activation through re-prompting, point-of-generation pre-confirmation through G.I.P., and post-hoc scrutiny through the audit protocol—are executed in each turn in this order, establishing multi-layered defense against drift that single-layer designs may fail to block.

Layer 1 (Re-declaration layer): Global rule reloading and persona application confirmation across all turns in Section I ["Output Management Section"]. By having the principal control rules explicitly re-declared at the head of each turn, attention to system-setting tokens—diminished by the progress of the context window—is actively re-activated. Placing this before the execution of G.I.P. means that G.I.P.'s three-condition confirmation acts on a more activated context state.

Layer 2 (G.I.P. layer): Confirmation of the three conditions (assumed user profile, persona, bias configuration) in Steps 1–3 by Section H ["Thought Verification Section"] in each turn. Receiving the re-activated state of Layer 1, the model linguistically self-attests the functional state of the three conditions and, in that state, generates and outputs the response, then hands it off to Layer 3.

Layer 3 (Self-Correction layer / audit protocol): Targeting the artifact handed off from Layer 2, the first phase of Section J ["End-of-Turn Execution Block"] performs self-audit (hereafter also referred to as Self-Correction) for violations of the persona and bias configuration. When a violation is detected, the relevant passage is revised and the revised description is output. Detailed conditions such as hallucination suppression, computational execution principles, and information-use stipulations are also confirmed and scrutinized at this layer.

In this design, Layer 1 (re-declaration) prepares the attention state that serves as the execution foundation for G.I.P.; Layer 2 (G.I.P.) secures the foundational conditions of generation at the point of generation; and Layer 3 (audit protocol) bears post-generation detailed scrutiny. With these three stages of control functioning in this order in each turn, statistical maintenance of constraint compliance in long-form sessions is achieved.

6.4 Observation of G.I.P.'s Effectiveness in Standalone Operation

A particularly noteworthy observation is that, in the investment analysis prompt, even when the self-audit function of Layer 3 (the first phase of Section J ["End-of-Turn Execution Block"]) was removed, G.I.P. alone (Section H ["Thought Verification Section"]) was confirmed to execute across all turns without drift in continuous analyses of at least eight turns.

This is positioned as empirical confirmation of this paper's theoretical claim that G.I.P. functions as a probabilistic improvement of constraint compliance not only in the context of multi-layered defense, but also as a standalone mechanism. The fact that maintenance of constraints in long-form sessions is achieved through pre-generation intervention alone—without dependence on the "post-hoc self-correction" critically examined by Huang et al. (2024)—demonstrates the originality of G.I.P.'s design philosophy.

6.5 Observation of Extensibility

As an observation independent of G.I.P.'s standalone effectiveness, the extensibility of the conditions incorporated into the pre-confirmation sequence was confirmed. By including, in addition to the standard three conditions of assumed user profile, persona, and bias configuration, condition items such as hallucination suppression (mandatory ⚠️ display when data is insufficient and mandatory explicit marking of inferences), computational execution principles (Python required), and prohibition of the use of social media and bulletin-board information, improved compliance with these conditions and suppression of drift were also confirmed.

The essential meaning of this observation is that G.I.P.'s effectiveness does not depend on the specific content of the standard three conditions. So long as a condition is incorporated into the pre-confirmation sequence, the mechanism functions—regardless of the content or type of that condition—by dynamically regenerating attention to the condition at the point of generation in each turn. Put differently, G.I.P. functions as a general-purpose protocol whose mechanism for improving condition compliance is content-agnostic.

This extensibility was likewise confirmed, as in Section 6.4, under the condition with the audit function removed. The observation that compliance with arbitrary condition items beyond the standard three improves through the pre-confirmation sequence handled by G.I.P. alone is positioned as an independent empirical record demonstrating the universal effectiveness of the pre-confirmation mechanism itself.

7. Comparison with Prior Work and Examination of Novelty

7.1 Comparison with Self-Correction Research

Kamoi et al. (2024) showed that the success conditions for self-correction require either "reliable external feedback" or "large-scale fine-tuning." G.I.P. uses neither. G.I.P. does not revise generated outputs post hoc; it linguistically confirms the functional state of conditions before generation. Its problem-setting therefore lies on a different dimension from the self-correction critically examined by Kamoi et al. (2024).

7.2 Comparison with Instruction Hierarchy Research

The instruction hierarchy research of Wallace et al. (2024) adopted the method of generating training data that prioritizes system prompts and fine-tuning GPT-3.5. G.I.P. requires no fine-tuning whatsoever.

"Reasoning Up the Instruction Ladder" (2025) shares a problem awareness with G.I.P. in the direction of inference about the instruction hierarchy before user-request execution as a meta-reasoning task, but adopts training-time intervention via the RLVR reinforcement

learning method. G.I.P. functions through prompt text alone, with minimal implementation cost.

7.3 Comparison with Constitutional AI

Constitutional AI (Bai et al., 2022) functions as a method that incorporates natural-language principle constraints into the model's training; at runtime, it operates with the learned principles internalized. G.I.P. functions without training-time intervention, through the runtime design of condition-confirmation syntax within the prompt alone. Furthermore, while CAI is designed as a post-hoc critique-and-revision cycle, G.I.P. is designed as pre-generation intervention at the point of generation.

7.4 Summary of G.I.P.'s Novelty

G.I.P.'s novelty can be summarized in three points.

First, the novelty of *extensibility*: compliance with conditions incorporated into the pre-confirmation sequence improves probabilistically regardless of the content or type of those conditions. This shows that G.I.P. does not depend on a specific constraint configuration (assumed user profile, persona, bias configuration), but rather functions through a universal effectiveness inherent to the pre-confirmation mechanism itself. Furthermore, this extensibility manifests even when the audit protocol is removed; the empirical confirmation of G.I.P.'s standalone improvement of condition compliance constitutes the empirical basis for this novelty.

Second, the novelty of the *design philosophy*: requiring no fine-tuning, training-time intervention, external feedback, or audit protocol, G.I.P. inserts condition re-activation at the point of generation through prompt text alone. The formalization—from the perspective of prompt design theory—of the fact that the "pre-generative, intrinsic approach"—self-attestation prior to generation, rather than post-hoc revision or audit—functions as an independent paradigm has not been found in prior work.

Third, the novelty of the *multi-layered defense architecture*: a design philosophy of three-layer division of roles—re-prompting (re-declaration) → G.I.P. (pre-confirmation) → audit protocol (post-hoc scrutiny). With each layer holding an independent function and functioning compositely in this order, more robust governance against drift—which single-layer designs may fail to block—is statistically realized as a composite design.

8. Applicability and Limitations

8.1 Applicability and Extensibility

Applications of G.I.P. include simultaneous maintenance of multiple independent constraint conditions in long-form sessions, long-term stabilization of persona configuration, multi-turn persistence of bias control, and improved compliance with hallucination-suppression stipulations.

A particularly noteworthy point as one of this paper's central claims is that the conditions G.I.P. confirms are not limited to the three conditions of assumed user profile, persona, and bias configuration. In the implementation observations on the investment analysis prompt, including condition items beyond these three—hallucination suppression, computational

execution principles, prohibition of social media information use, and so forth—in the pre-confirmation sequence likewise resulted in improved compliance with those conditions and suppression of drift.

That is, G.I.P. functions as a protocol extensible regardless of the content or type of conditions targeted for confirmation. Merely incorporating arbitrarily selected condition-confirmation items into the sequence yields the general effectiveness of improved compliance with those items. More importantly, this extensibility manifests even when the audit function (the post-hoc self-audit protocol) is removed. The observation that compliance with the included condition items improves through the pre-confirmation sequence handled by G.I.P. alone demonstrates the independent effectiveness inherent to the pre-confirmation mechanism itself.

This extensibility shows the universality of G.I.P.'s design philosophy and can be broadly applied as a design strategy that, beyond the specific use of investment analysis—incorporating into the pre-confirmation sequence any condition the prompt designer wishes to maintain (persona maintenance, stylistic stipulations, output format, ethical constraints, and so on)—probabilistically improves compliance with that condition in long-form sessions.

8.2 Limitations

The following limitations exist for G.I.P.

G.I.P. is a shift in the probability distribution effected by prompts and does not provide complete guarantees of constraint compliance. It cannot completely eliminate the influence of attention decay associated with context length described by Benderahmane et al. (2024); in extremely long sessions, its effectiveness may decline.

Because G.I.P.'s effects depend on linguistic execution of condition confirmation in each turn, there is the recursive challenge that the function declines if the G.I.P. construction itself drifts (for example, when the Step display is omitted). This recursive challenge—the problem that the pre-confirmation layer of G.I.P. itself is subject to the influence of attention decay—is positioned as an essential structural problem that cannot be resolved by single-layer measures.

The important answer to this problem is the design philosophy of securing robustness through compositing multi-layered drift-prevention functions. The implementation in the investment analysis prompt adopts a design in which, in addition to the G.I.P. layer (Section H ["Thought Verification Section"]), re-prompting (global rule re-declaration) by Section I ["Output Management Section"] is executed at the head of each turn. *Re-prompting* is a mechanism that, by having the principal control rules within the prompt explicitly re-declared at the head of each turn, actively re-activates attention to system-setting tokens diminished by the progress of the context window.

With three independent mechanisms—re-prompting (re-declaration), G.I.P. (pre-confirmation), and Self-Correction (post-hoc audit)—functioning compositely in this order in each turn, multi-layered defense against drift that single-layer designs may fail to block is established. In particular, by being placed at the head of each turn, re-prompting prepares the attention state that serves as the execution foundation for G.I.P., bears the complementary role of regenerating referential attention to system settings even when the G.I.P. construction itself has weakened, and functions as practical countermeasure against the recursive challenge. The session resumption detection mechanism (Section C ["Session

Resumption Detection"] functions as an ultimate safety net by monitoring G.I.P.'s execution state through the presence or absence of the "Drift Prevention  " marker and triggering re-execution of the entire procedure when absent. How this composite multi-layered architecture is to be systematized in general-purpose prompt design remains an issue for future work.

The empirical observations in this paper are qualitative; quantitative measurement of G.I.P.'s effects and systematic comparative verification across multiple models remain issues for future work.

9. Conclusion

This paper has proposed the **Governance Interlock Protocol (G.I.P.)** and systematically formalized its theoretical framework, its differentiation from existing methods, its design as a multi-layered defense architecture, and its condition confirmation sequence and flow. G.I.P. requires no fine-tuning, training-time intervention, or external feedback, and—through prompt text alone—inserts self-attestation of preconditions at the point of generation in each turn, achieving probabilistic improvement of constraint compliance.

The core finding this paper most emphasizes is the **extensibility** of G.I.P. In the implementation observations on the investment analysis prompt, in addition to the standard three conditions of assumed user profile, persona, and bias configuration, the inclusion of any other condition items in the pre-confirmation sequence likewise resulted in improved compliance with those conditions and suppression of drift. Furthermore, this effect manifests even when the post-hoc audit protocol is removed. That is, G.I.P. was empirically confirmed to function—even without an audit function—as a protocol whose included condition-confirmation items show improved compliance regardless of the content of those items, and which is therefore extensible without limit to the type of items. This shows that G.I.P.'s effectiveness derives not from the content of any specific three conditions, but from the universal effectiveness inherent to the pre-confirmation mechanism itself.

G.I.P.'s design philosophy is positioned as a paradigm shift to a "pre-generative, intrinsic approach" fundamentally distinct from: the limitations of post-hoc self-correction shown by Huang et al. (2024); the means of training-time intervention adopted by Wallace et al. (2024); and the post-hoc critique cycle designed by Constitutional AI (Bai et al., 2022). In the implementation in the investment analysis prompt, the three-layer division of roles—re-prompting (re-declaration) → G.I.P. (pre-confirmation) → audit protocol (post-hoc scrutiny)—functions compositely in each turn in this order, realizing a cooperative multi-layered defense architecture between pre-confirmation and post-hoc scrutiny. Through its extensibility, G.I.P. functions as a general-purpose design principle broadly applicable to any condition the prompt designer wishes to maintain.

This research, as a design theory of prompt engineering, presents a new control paradigm: pre-generative intervention at the point of generation. Future challenges include quantitative measurement of G.I.P. effects, comparative verification across multiple models, correlation analysis between the number of attestation steps and constraint compliance rates, drift prevention methods for the G.I.P. construction itself, and exploration of the optimal item composition for the pre-confirmation sequence.

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